

Professionally used software : Anaconda, Spider.

Text Book

T1. Ian Goodfellow, Yoshua Bengio, Aaron Courville, "Deep Learning", MIT Press, 2017

References

R 1. Duda, R.O., Hart, P.E., and Stork, D.G. Pattern Classification. Wiley-Inderscience, 2nd Edition. 2013

R2. Theodoridis, S. and Koutroumbas, K. Pattern Recognition. Edition 4, Academic Press, 2015

R3. Russell, S. and Norvig, N. Artificial Intelligence: A Modern Approach. Prentice Hall Series in Artificial Intelligence, 2013

R4. Bishop, C. M. Neural Networks for Pattern Recognition, Oxford University Press, 2008.

Weblinks:

W1: pu.informatics.global, <https://sm-nitk.vlabs.ac.in/>

Topics relevant to "SKILL DEVELOPMENT": Real time Data Analysis using Deep learning. Naming and coding convention for Data Science Project Development using ML/DL for Skill Development through Participative Learning techniques. This is attained through the Presentation as mentioned in the assessment component.

Course Code: CSE 313	Course Title: Storage Area Networks Type of Course: Theory Only Course	L- T-P- C	3 -0	0	3
Version No.	2.0				
Course Pre-requisites	Basics of information storage				
Anti-requisites					
Course Description	The course aims to equip students with basic introduction to Storage Area Networks, including storage architectures, logical and physical components of a storage infrastructure, managing and monitoring the data center and basic Disaster Recovery principles.				
Course Objective	The objective of the course is to familiarize the learners with the concepts of Storage Area Networks and attain Employability through Participative Learning techniques.				

Course Out Comes	On successful completion of the course the students shall be able to: CO1 Identify key challenges in managing information and analyze different storage networking technologies. [Understanding] CO2 Explain physical and logical components of a storage infrastructure of RAID, and intelligent storage systems. [Comprehension] CO3 Describe Object and Content addressed storage and storage virtualization. [Comprehension] CO4 Articulate business continuity solutions—backup and archive for managing fixed content. [Application]			
Course Content:				
Module 1	Storage System: Introduction to Information Storage	Assignment	Data Collection/Interpretation	10 Sessions
Topics: Information Storage, Evolution of Storage Architecture, Data Center Infrastructure, Virtualization and Cloud Computing. Data Center Environment: Application Database Management System (DBMS), Host (Compute), Connectivity, Storage, Disk Drive Components, Disk Drive Performance, Host Access to Data, Direct-Attached Storage, Data Proliferation				
Module 2	Data Protection – RAID, Intelligent Storage Systems	Case studies / Case let	Case studies / Case let	08 Sessions
Topics: RAID Implementation Methods, RAID Array Components, RAID Techniques, RAID Levels, RAID Impact on Disk Performance, RAID vs SSD, Types of RAID Storage for Databases in Public Cloud Intelligent Storage Systems: Components of an Intelligent Storage System, Types of Intelligent Storage Systems, Optimal architectures for intelligent storage systems				
Module 3	Object-Based and Unified Storage	Quiz	Case studies / Case let	08 Sessions
Topics: Object-Based Storage Architecture: Components of OSD, Object Storage and Retrieval in OSD, Benefits of Object-Based Storage, Content-Addressed Storage. Virtualization in SAN: types of storage virtualization, Benefits of virtualization				
Module 4	Backup and Archive, Replication	Quiz	Case studies / Case let	10 Sessions
Backup Purpose, Backup Considerations, Backup Granularity, Data Recovery Services, Backup Methods, Backup Architecture, Backup and Restore Operations, Backup Topologies, Backup in NAS Environments.				

Local Replication: Replication Terminology, Uses of Local Replicas, Replica Consistency, Local Replication Technologies, Tracking Changes to Source and Replica, Restore and Restart Considerations, Creating Multiple Replicas.
Remote Replication: Modes of Remote Replication, Remote Replication Technologies.
Targeted Application & Tools that can be used: Architecture based environment
Text Book T1. G. Somasundaram, Alok Shrivastava. "Information Storage and Management", EMC Education Services, Wiley India. 2nd Edition.2012.
References R1. Ulf Troppens, Rainer Erkens and Wolfgang Muller. "Storage Networks Explained", Wiley India. 2nd Edition.2015. R2. Rebert Spalding. "Storage Networks The Complete Reference", Tata McGraw Hill, Indian Edition.2017. R3. Richard Barker and Paul Massiglia. "Storage Area Networks Essentials A Complete Guide to Understanding and Implementing SANs", Wiley. 1stEdition.2008. E-Resource: https://presiuniv.knimbus.com/user#/home R3 Web resources: Students may find articles and significance of SAN at https://www.ibm.com/topics/storage-area-network and EMC2 and may refer an eBook on "Storage Area Network Essentials" A Complete Guide to Understanding and Implementing SANs by Richard Barker, Paul Massiglia https://presiuniv.knimbus.com/user#/searchresult
Topics relevant to "EMPLOYABILITY SKILLS": Data Protection – RAID for developing Employability Skills through Participative Learning techniques. This is attained through assessment component mentioned in course handout.